Early Detection of Temporal Anomalies in B2B Procurement Using Customer–Part Purchase Patterns

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***Abstract*—Temporal patterns exist in repeated procurement transactions, whose detection can be helpful to find unusual replenishment transactions that occur early. But, the purchase intervals vary widely for each customer-part, so a single interval threshold will not work globally. This paper proposes a temporal screening pipeline with customers and each item that essentially computes inter-purchase intervals (IPIs), builds past-only historical and recent statistics, and pinpoints IPIs that are shorter than expected based on the purchase history of that pair of customers and items. The membership of an alert is based on a median based threshold, while a median based absolute deviation (MAD) score is used for ranking; the unsupervised benchmarks Isolation Forest and Local Outlier Factor (LOF) are compared. A total of 64,892 events after meeting the minimum-history requirement and 136,372 temporal events were recovered from the 285,941 records in six annual procurement files. 13,861 events (21.36%), or 30% relative to the historical median, were flagged on the complete eligible population, while 19,983 events (30.79%) were flagged at 50% relative to the historical median. At an injection ratio of 30% the 30% median rule identified 100% of the injected cases with controlled precision of 83.78%, at a chronological holdout the injected cases were not detected at all. The median – MAD rule, Isolation Forest and LOF give complementary comparisons but none of them can be considered as real-world anomaly accuracy, due to the lack of confirmed anomaly labels. The prototype that comes as a result of this gives an audit trail to review the decision rather than an automatic recommendation for unnecessary purchasing.**

***Index Terms*—B2B procurement, temporal anomaly detection, inter-purchase interval, robust statistics, median absolute deviation, Isolation Forest, Local Outlier Factor, supply-chain analytics.**

I. Introduction

Procurement systems generate repeated transactions for customers and parts over time. These records can support more than demand-quantity forecasting: the timing between consecutive purchases can also reveal changes in replenishment behaviour. Existing research has examined intermittent spare-part demand, procurement decision-support, supply-chain risk, purchasing anomalies, and customer purchase behaviour [1]–[5]. Yet these problem settings do not necessarily identify an unusually early reorder for a specific customer–part pair.

A global rule such as "reorder within k days" can be misleading because one part may be purchased daily while another is purchased only a few times each year. This paper therefore models the interval between successive purchase dates within each customer–part combination and compares the current interval with that pair's own historical behaviour. The resulting signal is designed to prioritize events for investigation, not to assert that an order is erroneous or wasteful.

The study makes four contributions:

- It represents early reorder screening as a customer–part-time problem with inter-purchase interval (IPIs).
- It makes historical and recent features based on only intervals that occur before the event that is being scored.
- It investigates the threshold sensitivity and compares the median rule and median–MAD detector with Isolation Forest and LOF with chronological holdout and controlled synthetic cases.
- It offers a reproducible Python pipeline and a Streamlit dashboard to rank and explore the candidate events.

The primary research questions are: (1) can such a rule, specific to the customer, offer an interpretable screening signal; (2) what impact does the threshold have on the amount of reviews; and (3) how does this rule compare with statistical and unsupervised approaches in a controlled synthetic assessment? Not having the ability to have anomaly labels verified by experts constrains statements concerning actual operational performance.

II. Related Work

A. Spare-Part Demand Forecasting

The field of spare parts procurement has been subject of numerous research studies focusing on intermittent demand, life cycle effects, data pooling, and feature engineering. Kenaka *et al.* explore an ensemble method that combines focal loss and SMOTE technique for forecasting intermittent demand for spare parts [1]. Goldsmith and Sachs analyse the data-pooling scheme derived from the Bass life-cycle model [2] and Kunić *et al.* discuss the feature engineering and ABC–XYZ segmentation [6]. There is some work that deals with long-term demand for spare parts in end-of-life product scenarios and the contribution of feature engineering compared to model complexity [7], [8]. Intermittent spares forecasting has also been explored using LLMs and other machine-learning techniques [3], [9]. These studies serve as bases for the treatment of sparse and irregular purchasing data; however, the prime target of these studies is the quantity of demand or forecasting performance and not the unusually short time period of one customer–part pair.

B. Procurement Analytics and Anomaly Detection

Machine learning techniques have been used for supplier selection, order allocation, and purchase decision support [4], [10]. Another line of research in public-procurement has tackled the issue of fraud and corruption detection [11], [12]. Herreros-Martínez *et al.* propose a two-stage clustering and isolation forest method for detecting anomalies in enterprise purchase processes [13]. The analytical methods applied to procurement transactions in these works are similar to the present study but the definition of anomaly is not as broad as in this paper: In this case, the event is screened because the current IPI is unusually short compared with the same customer–part history. Related operational risks are covered by supplier-delay warning and supply-chain disruption prediction, with the former trying to tackle the actions of the supplier and the latter focusing on the level of the supply chain. [14]–[18].

C. Customer Purchase Behaviour and Research Gap

Customer loyalty, customer lifetime value, consumer analytics, buying-habit prediction, and next-purchase timing have been studied in the context of purchase behaviour [19]–[24]. Additional research investigates the B2B lead prioritization problem and the purchase/churn prediction problem [25], [26]. The use of AI in procurement and supply management is reported in various reviews and shows a wide variety of applications and ongoing research opportunities [4], [5]. The gap that has been covered in the present work is not the general anomaly detection or purchase prediction but rather the explicit screening of the short intervals between purchases in the history of a single customer. The literature reviewed supports this formulation, but does not prove that any previous research has attempted to solve a problem similar to this one.

D. National and International Current Status

Digital procurement platforms are now increasingly using analytics on the procurement records in India. AI and machine-learning functionalities have been reported to be used on the Government e-Marketplace (GeM) to detect any possible procurement patterns of suspected cartelisation, collusion or splitting of orders [27]. Furthermore, studies with Indian manufacturing companies have explored the linkages between AI, big-data analytics, supply-chain flexibility and firms' performance [28]. Globally, AI and machine learning systems have been reviewed in various aspects of purchasing and supply management, such as forecasting, supplier selection, order management, identification of anomalies, and fraud detection [4], [5]. These projects are the foundation for the broader context of the data-driven procurement analytics. The present work is focused on a more limited objective: to identify unusually short reorder intervals for each customer–part instead of identifying procurement fraud or predicting the quantity of the demand.

III. Problem Formulation

Let a customer–part pair be denoted by $g = (c, p)$, where c is a customer identifier and p is a part identifier. After sorting distinct purchase dates for pair g, let $t_{g,i}$ be the date of its ith purchase. The inter-purchase interval is

$$IPI_{g,i} = t_{g,i} - t_{g,i-1}, \tag{1}$$

measured in days. Duplicate records for the same customer, part, and date are collapsed for interval construction so that they do not create zero-length intervals.

For an event i, let $\mathcal{H}_{g,i}$ contain only intervals preceding $IPI_{g,i}$. Define the historical median and median absolute deviation as

$$M_{g,i} = \text{median}(\mathcal{H}_{g,i}), \tag{2}$$

$$MAD_{g,i} = \text{median}_{x \in \mathcal{H}_{g,i}} |x - M_{g,i}|. \tag{3}$$

An event is eligible only when at least three previous intervals are available, its current IPI and historical statistics are valid, and $M_{g,i} > 0$. The main screening rule flags an event when

$$IPI_{g,i} \leq \tau M_{g,i}, \tag{4}$$

where $\tau = 0.30$ is the selected operating threshold. A flagged event is a candidate for review; it is not a verified anomaly or evidence of over-ordering.

IV. Dataset and Preprocessing

The dataset in question is the industrial procurement transactions dataset B2B-Parts-Rec hosted on Zenodo, which is a real, anonymized dataset (DOI: 10.5281/zenodo.19492687) [29]. The Zenodo description lists 219 industrial client companies in the food and beverage manufacturing sector, but the 6 provided files contain 238 different customer identifiers. Records are provided for industrial spare parts, industrial machinery parts, and catalog based purchases made on several continents (including Africa, Asia, Australia, Europe and

TABLE I
DATASET AND TEMPORAL FEATURE CONSTRUCTION

Measure	Count / setting
Raw procurement records	285,941
Temporal purchase events	136,372
Customer–part groups	34,664
Events with sufficient history	64,892
Minimum prior intervals	3
Historical reference	Expanding median and MAD
Recent reference	Median of up to 3 prior intervals

North America) [29]. The project uses six files of the form of semicolon (';')-delimited data, one for each year from 2024 to 2029. The 2027–2029 dates are future dated relative to the study period, and are not necessarily calendar dates but may be dataset provided and shifted or synthetic dates and should be interpreted as such; the evaluation relies on their relative chronological order. The pipeline is designed to handle the reading of customer identifier, part identifier and request date for temporal analysis. There are 285,941 raw records in the combined input. Records that do not include all of the temporal keys are deleted, purchase date is split up and distinct combinations of customer, part and date are sorted chronologically. Current IPI is calculated from the subsequent two different dates in the pair.

In each event, the historical feature builder moves the sequence of intervals before computing the expanding median, expanding MAD and recent median of the last 3 intervals. This is a key change: the event being assessed is not permitted to add to its own historical reference statistics. At least three previous intervals are needed for screening to be ready for model.

The 64,892 model-ready events represent the eligible population used for the final threshold comparison and detector experiments. Table I summarizes the main processing counts and the temporal features used. The reduction from raw records to temporal events reflects the construction of distinct customer–part purchase dates, while the eligible population includes only events with at least three prior intervals. The raw record count, temporal event count, and eligible event count refer to different processing stages and should not be treated as interchangeable denominators.

V. PROPOSED METHODOLOGY

A. Past-Only Temporal Features

In addition to $IPI_{g,i}$, the pipeline computes the ratio of the current interval to the historical median,

$$R_{g,i} = \frac{IPI_{g,i}}{M_{g,i}}, \tag{5}$$

and a recent interval ratio using the median of the previous three intervals when available. The difference $M_{g,i} - IPI_{g,i}$ expresses the number of days by which the current interval is shorter than the historical median. The history count is retained as context, while customer and part identifiers are used to form groups rather than as numeric detector features.

B. Threshold-Based Candidate Detection

The main detector uses (4). A threshold sweep evaluates $\tau \in \{0.50, 0.40, 0.30, 0.20\}$ to quantify how strictness affects alert volume. The 30% rule is selected as the current operating point for the prototype. This is a design choice based on the alert-volume trade-off and controlled experiments, not a universally optimal or clinically calibrated cut-off.

C. Robust Ranking and Severity

To support investigation, the prototype calculates an earlyness component

$$E_{g,i} = \max(0, \min(1, 1 - R_{g,i})), \tag{6}$$

and a robust deviation component. To prevent division by zero, the robust scale is floored at one day:

$$S_{g,i} = \max(1.4826\, MAD_{g,i}, 1). \tag{7}$$

The robust early score is $Q_{g,i} = \max(0, (M_{g,i} - IPI_{g,i})/S_{g,i})$. The dashboard's heuristic ranking score combines earlyness with a capped log-transformed robust component:

$$Z_{g,i} = 0.70E_{g,i} + 0.30\frac{\log(1 + Q_{g,i})}{1 + \log(1 + Q_{g,i})}. \tag{8}$$

This score is a ranking heuristic, not a probability. Severity labels are assigned from the interval ratio: very high for $R \leq 0.10$, high for $0.10 < R \leq 0.20$, and moderate for $0.20 < R \leq 0.30$. These labels describe rule-based severity bands, not validated business outcomes.

Fig. 1 summarizes the system architecture. First, the six yearly procurement files are loaded and cleaned, and distinct purchase dates are grouped by customer–part pair. Second, the pipeline derives each inter-purchase interval and computes historical median, MAD, and recent-interval statistics using only earlier events, preventing the event under review from influencing its own baseline. Third, the 30% historical-median rule identifies candidate early reorders, while the ranking heuristic orders candidates for inspection rather than assigning anomaly probabilities. Fourth, the Streamlit dashboard exposes the alert list, filters, event details, and the pair's purchase history to support human review; it does not automatically change orders. Finally, a separate offline evaluation compares threshold settings and alternative detectors using a chronological holdout and controlled synthetic early-interval cases. This separation keeps operational screening distinct from experimental evaluation.

Fig. 2 gives an event-level example of the core idea: the current interval is compared with the historical intervals available before that event for the same customer–part pair. When the current interval is substantially shorter than its historical median, it satisfies the early-reorder rule and can be flagged for review. The historical baseline is pair-specific, so the method does not assume that all parts should be reordered at the same frequency.

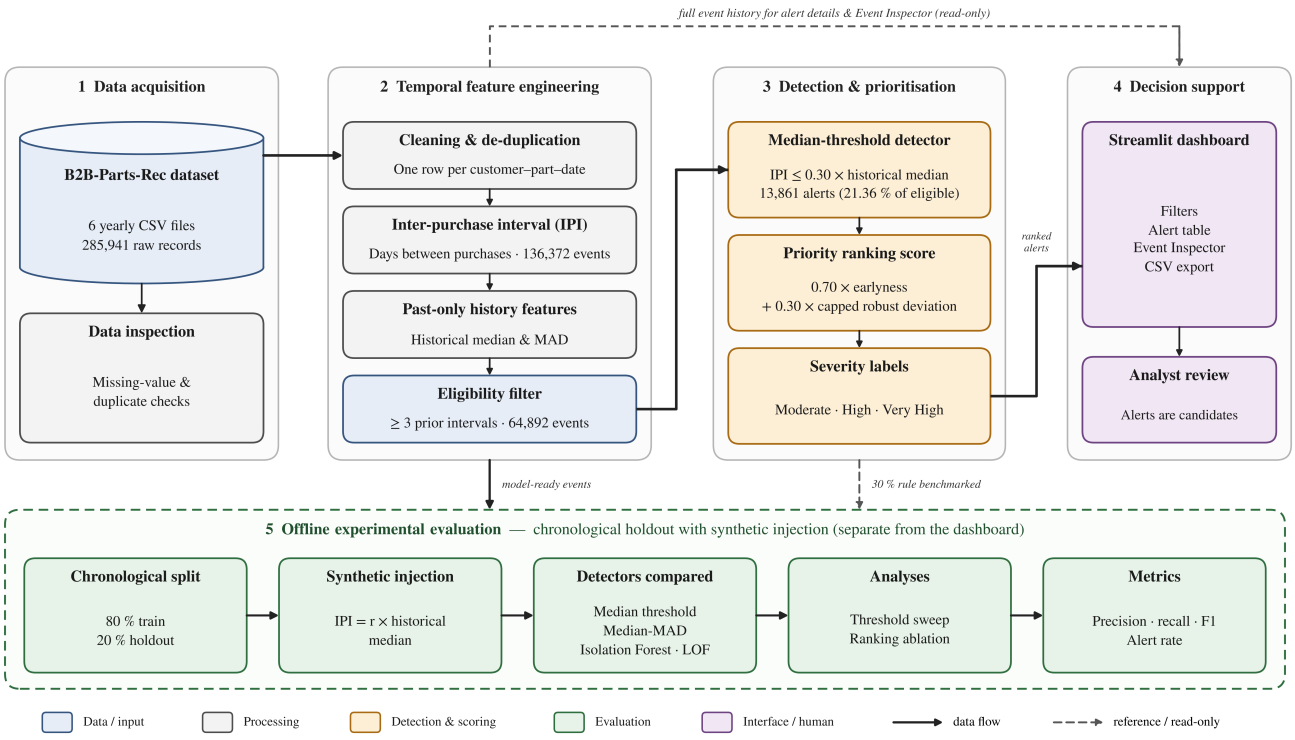


Fig. 1. System architecture of the proposed customer–part temporal screening pipeline, covering data acquisition, temporal feature engineering, early-reorder detection, dashboard-based decision support, and offline experimental evaluation.

D. Prototype Workflow

Implementation – Python based data pipeline, a Streamlit dashboard. Features alert ranking, filtering, event-level inspection and historical context for selected customer–part combinations. The dashboard is designed to assist a human reviewer to review candidate events. Does not automatically cancel orders or imply that an order is not needed.

VI. Experimental Design

A. Chronological Holdout

The detector comparison sorts eligible events based on the request date and the first 80 (51,913 events) is used as the training partition with the last 20 (12,979 events) as the natural holdout. Medians in feature imputation for Isolation Forest and LOF are calculated from training partition. In this setting, a chronological split would be better than a random split, as it would approximate the scoring of later events, after earlier events.

B. Controlled Synthetic Cases

There are no labels in the transaction data that confirm any expert that are related to authentic early-reorder anomalies. To assess the sensitivity at a known construction, the simulated events of a natural holdout are duplicated and their current IPI is modified by a chosen fraction of the historical median: 0.50, 0.30, 0.20, or 0.10. The historical features and training data remain constant. Positive label is the synthetic copy and reference group is the unmodified holdout. Reference events don't prove they are normal and injected cases aren't real observed anomalies. So, the only synthetic-evaluation metrics that control precision, recall, and F1 in this setup are those of the synthetic evaluations.

C. Compared Methods and Metrics

The comparisons are performed with (i) median-threshold rules at various ratios, (ii) a strong median–MAD deviation rule at various strong score thresholds, (iii) the Isolation Forest, and (iv) the Local Outlier Factor. The reported thresholds for Isolation Forest and LOF are derived from training anomaly scores and not a complete hyperparameter search. Both detector comparison and controlled precision reports synthetic detection rate and natural alert rate, while both detector comparison and ablation reports Recall@150. These are not actual measurements of anomaly accuracy but are synthetic evaluations, which are controlled.

VII. Results and Comparative Analysis

A. Threshold Sensitivity

Table II summarizes the threshold sweep on the complete eligible population of 64,892 events; its alert rates, including 21.36% at the 30% threshold, use this full eligible set as the denominator. As the threshold becomes stricter, fewer events meet the rule: the 50% setting flags the most events, while the 20% setting flags the fewest. The table therefore shows the trade-off between a broader screening net and a smaller review workload. The 30% rule flags 13,861 of 64,892 eligible

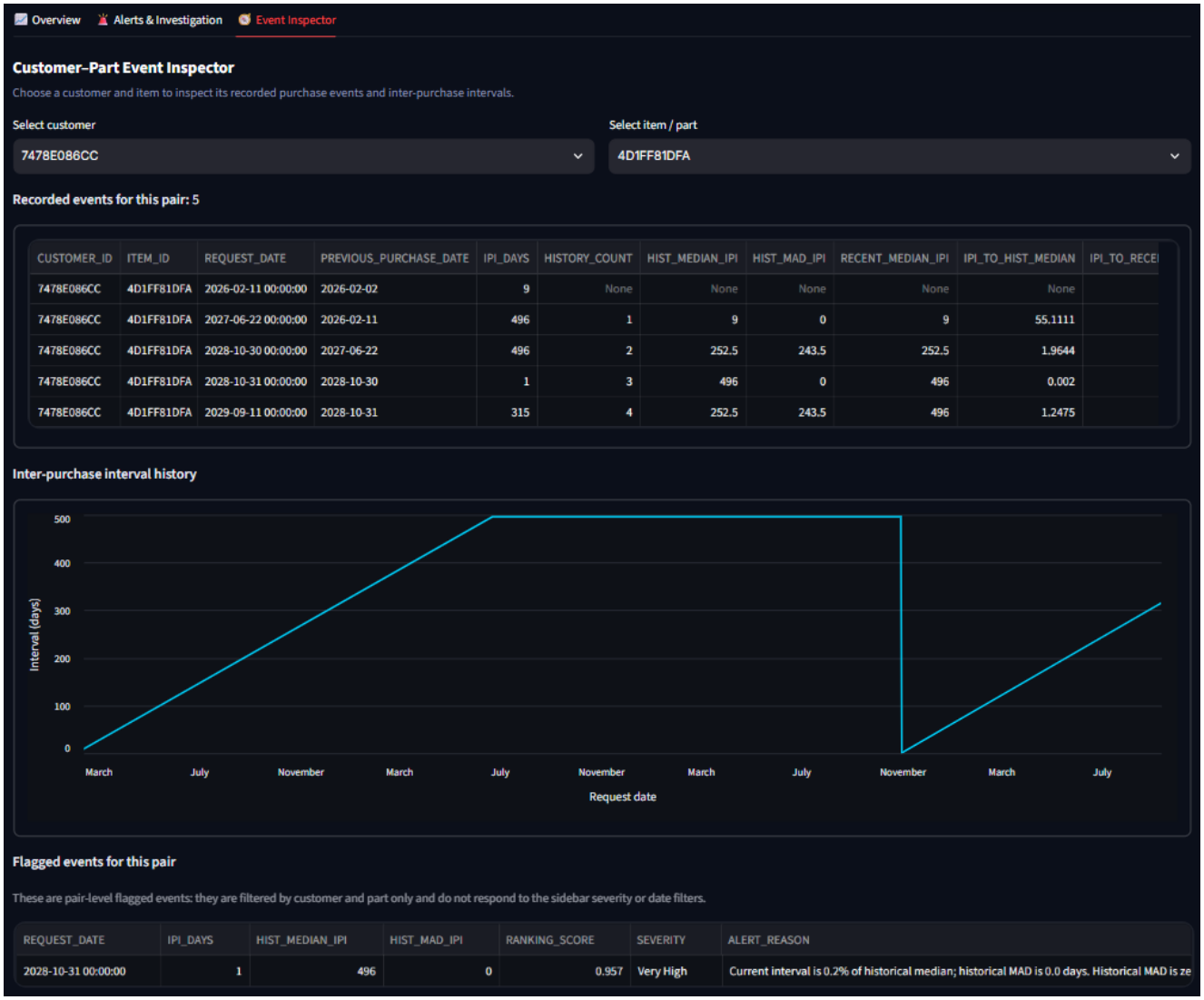


Fig. 2. Event-level illustration of the current purchase interval and its comparison with past-only history for the same customer–part pair.

TABLE II
THRESHOLD SENSITIVITY ON THE COMPLETE ELIGIBLE POPULATION

Threshold	Flagged	Alert rate	Reduction vs. 50%
50%	19,983	30.79%	0.00%
40%	17,217	26.53%	13.84%
30%	13,861	21.36%	30.64%
20%	10,234	15.77%	48.79%

events (21.36%). Relative to the 50% rule, it flags 6,122 fewer events, a 30.64% reduction in alert volume. This is a reduction in the number of events sent for review; it is not evidence that 30.64% of false positives were removed.

B. Detector Comparison

The results of the detectors reported in Table III are controlled results from the last detector-comparison report from the repository. For the 30% injection setting that was tested, the median threshold has a 100% synthetic detection rate and a controlled precision of 83.78%. At the stricter settings, the median-MAD rules have lower detection rates than in this tested setting, and the rates of Isolation Forest and LOF are lower. These three measures are reflected in figure 3 which illustrates them for the compared methods, providing a clearer way to compare these trade-offs. The 19.35% natural alert rate for the 30% rule is computed on the chronological holdout events (12,979), but the 21.36% is computed on the full eligible population (64,892). The lower natural alert rate of a detector should not be interpreted as better real-world precision as the lower natural group has not been confirmed as being normal. Importantly, the synthetic test includes cases built at exactly the same ratio as the evaluated test that was used in the historical data, namely 30% – near perfect detection at that ratio is thus predicted by construction, and should therefore be viewed as a consistency check and not as a stand-alone evidence of generalisation.

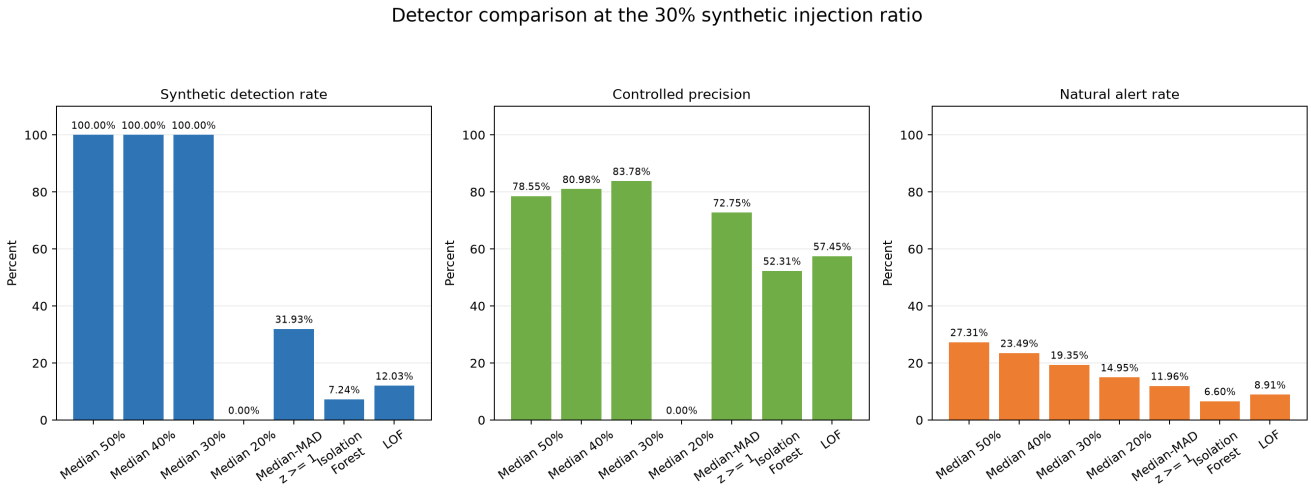


Fig. 3. Controlled detector metrics at a 30% injection ratio. Synthetic detection and controlled precision are measured on constructed cases; natural alert rate is measured on the chronological holdout.

TABLE III
CONTROLLED DETECTOR COMPARISON AT 30% INJECTION RATIO

Method	Detection	Precision	Natural alert rate
Median threshold, 50%	100.00%	78.55%	27.31%
Median threshold, 40%	100.00%	80.98%	23.49%
Median threshold, 30%	100.00%	83.78%	19.35%
Median threshold, 20%	0.00%	0.00%	14.95%
Median–MAD, $z \geq 1$	31.93%	72.75%	11.96%
Median–MAD, $z \geq 2$	13.48%	73.28%	4.92%
Median–MAD, $z \geq 3$	8.67%	72.67%	3.26%
Median–MAD, $z \geq 4$	6.66%	71.96%	2.60%
Isolation Forest	7.24%	52.31%	6.60%
LOF	12.03%	57.45%	8.91%

TABLE IV
RANKING ABLATION AT A 30% INJECTION RATIO

Ranking method	Detection rate	Recall@150
Threshold only	100.00%	0.00%
Earlyness only	100.00%	0.00%
Robust only	100.00%	0.92%
Combined 70/30	100.00%	0.00%

C. Ranking Ablation

The four choices of ranking are compared in Table IV while keeping the membership of the events within the alert set constant at the 30% threshold and only varying the order by which flagged events are ranked. Detection rate tells whether or not injected cases are marked; Recall@150 tells how many injected cases are in the top 150 ranked alerts. The same overall percentage is detected by all methods in this experiment, while only robust-only ranking adds any injected cases to the top 150. As a result, each variant of the rankings will be equal in terms of alert membership and natural alert rate. The earlyness-only ranking is monotonic with the interval ratio and does not contribute anything more than threshold ranking. Recall at most 1.16% (with 150 retrieval events out of 12,979 retrieval events in the holdout) is not a large amount of prioritization gain as it is only 0.92% of recall for robust-only ranking. The existing 70/30 combined score is no better than strong-only ranking. Thus, there is no evidence that the composite score is better.

The ablation result suggests that the threshold is responsible for alert membership, while robust-only ranking retrieves a small fraction of injected cases in the top 150. This result is limited and specific to the synthetic test design. The combined score remains a heuristic retained for dashboard ordering, not a validated model improvement.

D. Dashboard

The dashboard features an operational interface to explore flagged events, filter the candidate list and examine a flagged event in conjunction with the corresponding customer–part history. The dashboard overview is displayed in Fig. 4 and includes the number of eligible events and alerts, alert distribution summaries and navigation to view the candidate events. This implementation shows how the screening output could be made reviewable and explainable at event level.

VIII. DISCUSSION, LIMITATIONS, AND SUSTAINABILITY RELEVANCE

The experiments demonstrate that a simple customer-part-specific median rule is understandable, and can identify most of the injected early intervals at an injection ratio of 30%. It also provides direct control of the volume of reviews, with lower volumes of events being flagged if the thresholds are tightened. Comparisons of the selected rule with the median and the MAD, Isolation Forest, and LOF give context to the selected rule, but the evidence is not enough to confirm universal superiority or production readiness.

There are a number of limitations, which are material. The dataset does not include any known labels for true early-

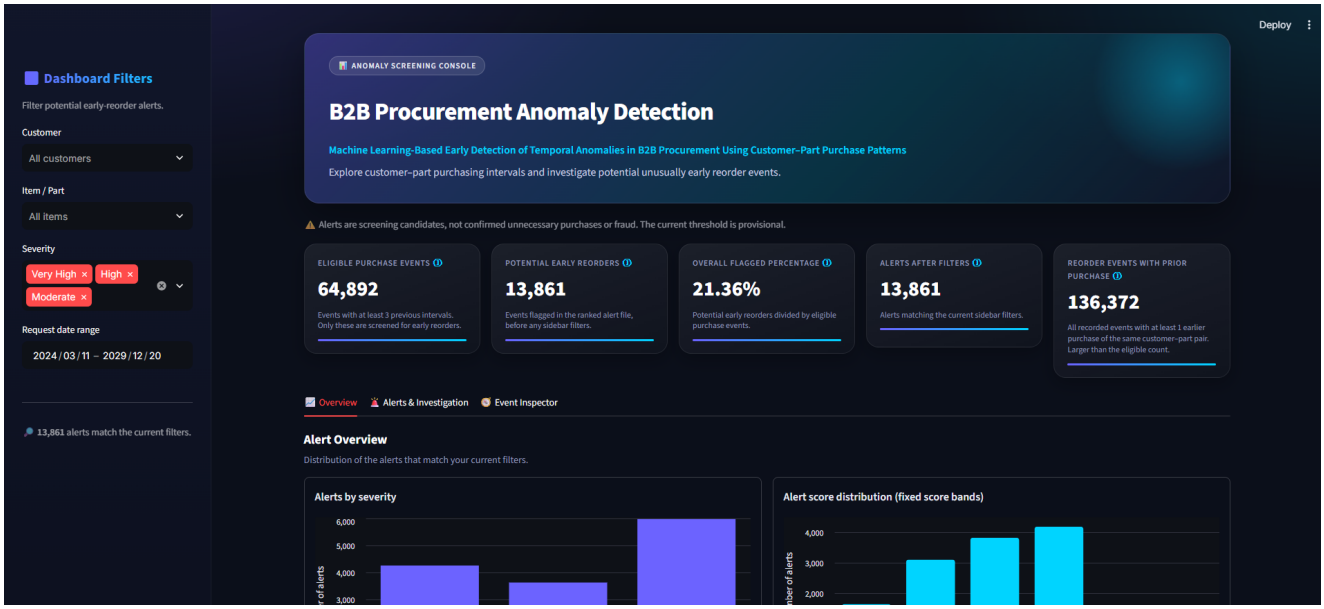


Fig. 4. Streamlit dashboard for reviewing candidate early-reorder events, including eligible-event and alert counts, alert distributions, and navigation for event-level investigation.

reorder anomalies, so the precision, recall and false-positive rates are not known in the real world. Second, synthetic cases are produced by replacing the present IPIs with fractions of a set of historical medians; the test case is applied to the detector as another mathematical construction, but may not represent all operational reasons for early purchase. Third, there is only one chronological division reported. Fourth, the existing 70/30 weightage assigned to the two components of the ranking is heuristic, and the ablation does not demonstrate that the combination of the two is superior to robust-only ranking. Furthermore the study does not make a comparison with a fixed day threshold, which means that the motivation for customer-specific thresholds is conceptually argued rather than empirically validated on the basis of that threshold. Lastly, the project has not been able to measure interventions, procurement savings, inventory reductions, waste or emissions.

The prototype could be useful for the United Nations Sustainable Development Goal (SDG) 12, in that it creates a repeatable signal that can be used as a basis for analysing purchasing and replenishment patterns, and to SDG 9 as an instance of data-driven process analytics. These are guidelines and may not include all items. The reduction of alerts observed is not a measure of reduced waste or reduced use of resources and there is not a claim to quantified sustainability impact.

IX. Conclusion and Future Work

This paper presented a customer–part-specific temporal screening pipeline for identifying unusually short inter-purchase intervals in B2B procurement records. The pipeline uses past-only historical median, MAD, recent interval statistics, a 30% historical-median rule, and a dashboard-oriented ranking heuristic. Across 285,941 raw records, it produced 136,372 temporal events and 64,892 eligible events. The 30% rule flagged 13,861 eligible events (21.36%), 30.64% fewer alerts than the 50% rule on the same population. In a controlled synthetic evaluation, the 30% rule detected 100% of injected cases at the 30% injection ratio with 83.78% controlled precision. These metrics do not represent real-world accuracy because the natural reference group is not verified normal and the positive cases are synthetic.

Future work should include domain-expert review of candidate alerts, comparison with a global fixed-day reorder baseline, multiple rolling chronological evaluations, testing alternative minimum-history requirements, and validating whether alerts correspond to actionable procurement issues. Field evaluation with outcome data would be required before claiming reductions in costs, over-ordering, waste, or emissions.

References

[1] S. P. Kenaka, A. Cakravastia, A. Ma'ruf, and R. T. Cahyono, "Enhancing Intermittent Spare Part Demand Forecasting: A Novel Ensemble Approach with Focal Loss and SMOTE," *Logistics*, vol. 9, no. 1, p. 25, 2025, doi: 10.3390/logistics9010025.

[2] R. L. Goldsmith and A.-L. Sachs, "Spare Part Demand Forecasting in Every Phase: A Data Pooling Approach to the Bass Life Cycle Model," *International Journal of Production Research*, 2025, doi: 10.1080/00207543.2025.2607643.

[3] P. G. Giannopoulos, T. K. Dasaklis, I. Tsantilis, and C. Patsakis, "Machine Learning Algorithms in Intermittent Demand Forecasting: A Review," *International Journal of Production Research*, 2025, doi: 10.1080/00207543.2025.2578701.

[4] D. Balkan and G. A. Akyuz, "Artificial Intelligence (AI) and Machine Learning (ML) in Procurement and Purchasing Decision-Support (DS): A Taxonomic Literature Review and Research Opportunities," *Artificial Intelligence Review*, vol. 58, p. 341, 2025, doi: 10.1007/s10462-025-11336-1.

[5] J. M. Spreitzenbarth, C. Bode, and H. Stuckenschmidt, "Artificial Intelligence and Machine Learning in Purchasing and Supply Management: A Mixed-Methods Review of the State-of-the-Art in Literature and Practice," *Journal of Purchasing and Supply Management*, vol. 30, no. 1, p. 100896, 2024, doi: 10.1016/j.pursup.2024.100896.

[6] Z. Kunić, V. Dakić, A. Radovan, and F. Furko, "Derived Feature Engineering and ABC–XYZ Segmentation for Machine Learning-Based Forecasting of Intermittent Spare Parts Demand," *Applied Sciences*, vol. 16, no. 12, p. 5804, 2026, doi: 10.3390/app16125804.

[7] S. Nathan, B. V. S. Reddy, V. S. S. Sujan, C. C. Sastry, J. Krishnaiah, and S. Jitpichitchai, "A Machine Learning Framework for Long-Term Forecasting of Spare Part Demand in End-of-Life Product Scenarios," *Scientific Reports*, vol. 16, p. 1394, 2026, doi: 10.1038/s41598-025-31171-2.

[8] B. S. Nathan, P. M. Aravinth, B. V. S. Reddy, C. C. Sastry, S. Salunkhe, and R. Cep, "Primacy of Feature Engineering Over Architectural Complexity for Intermittent Demand Forecasting," *Scientific Reports*, vol. 16, p. 4792, 2026, doi: 10.1038/s41598-026-35197-y.

[9] M. Abbaszadeh Nakhost, A. Bloemer, and S. Minner, "Full Automation or Just a Copilot? A Study of Large Language Models on Spare Parts Demand Forecasting with OpenAI o1-mini," *International Journal of Production Research*, 2025, doi: 10.1080/00207543.2025.2591225.

[10] E. N. Hayati, W. A. Jauhari, R. W. Damayanti, and C. N. Rosyidi, "An integrated analytics model for supplier selection and order allocation with machine learning and multi-criteria optimization," *Decision Analytics Journal*, vol. 16, art. no. 100599, 2025, doi: 10.1016/j.dajour.2025.100599.

[11] E. Schneider dos Santos, M. Machado dos Santos, M. Castro, *et al.*, "Detection of Fraud in Public Procurement Using Data-Driven Methods: A Systematic Mapping Study," *EPJ Data Science*, vol. 14, p. 52, 2025, doi: 10.1140/epjds/s13688-025-00569-3.

[12] M. Ndolo, A. Wanjoya, and P. Kasyoka, "A Hybrid Machine Learning Model for Detecting and Preventing Corruption in Kenya's Public Procurement Contracts," *Machine Learning Research*, vol. 10, no. 2, pp. 131–136, 2025, doi: 10.11648/j.mlr.20251002.14.

[13] A. Herreros-Martínez, R. Magdalena-Benedicto, J. Vila-Francés, A. J. Serrano-López, S. Pérez-Díaz, and J. J. Martínez-Herráiz, "Applied Machine Learning to Anomaly Detection in Enterprise Purchase Processes: A Hybrid Approach Using Clustering and Isolation Forest," *Information*, vol. 16, no. 3, p. 177, 2025, doi: 10.3390/info16030177.

[14] A. Alaoua and M. Karim, "Intelligent Early Warning System for Supplier Delays Using Dynamic IoT-Calibrated Probabilistic Modeling in Smart Engineer-to-Order Supply Chains," *Applied System Innovation*, vol. 8, no. 5, p. 124, 2025, doi: 10.3390/asi8050124.

[15] C. Xiaoyang, "Explainable Dual LSTM-Autoencoders with Exogenous Features for Anomaly Detection and Supply Chain Forecasting," *Scientific Reports*, vol. 15, p. 42371, 2025, doi: 10.1038/s41598-025-26449-4.

[16] M. A. Khaled, M. Nadeem, A. G. E. H. A. Mansour, A. A. M. Kulfod, and M. A. Aziz, "Machine Learning Approaches for Early Prediction of Supply-Chain Disruptions and Operational Risks," *Journal of Information Systems Engineering and Management*, vol. 9, no. 4s, pp. 4137–4148, 2024, doi: 10.52783/jisem.v9i4s.14812.

[17] M. Wyrembek, G. Baryannis, and A. Brintrup, "Causal Machine Learning for Supply Chain Risk Prediction and Intervention Planning," *International Journal of Production Research*, vol. 63, no. 15, pp. 5629–5648, 2025, doi: 10.1080/00207543.2025.2458121.

[18] H. Xia, S. Sani, J. Milisavljevic-Syed, and K. Salonitis, "Integrated Machine Learning Model for Managing Customer-Driven End-to-End Supply Chain Uncertainty," *International Journal of Production Research*, 2026, doi: 10.1080/00207543.2026.2648764.

[19] S. Roosta, S. J. Sadjadi, and A. Makui, "Predicting Customer Loyalty in Omnichannel Retailing Using Purchase Behavior, Socio-Cultural Factors, and Learning Techniques," *PLOS ONE*, vol. 20, no. 8, p. e0330338, 2025, doi: 10.1371/journal.pone.0330338.

[20] A. Wong, A. Viloria Garcia, and Y.-W. Lim, "A Data-Driven Approach to Customer Lifetime Value Prediction Using Probability and Machine Learning Models," *Decision Analytics Journal*, vol. 16, p. 100601, 2025, doi: 10.1016/j.dajour.2025.100601.

[21] T. Stylianou and A. Pantelidou, "A Machine Learning Approach to Consumer Behavior in Supermarket Analytics," *Decision Analytics Journal*, vol. 16, p. 100600, 2025, doi: 10.1016/j.dajour.2025.100600.

[22] Z. Wang, "The Application of Deep Learning in User Purchase Behaviour Analysis and Marketing Decision Support," *International Journal of Computational Intelligence Systems*, vol. 18, p. 253, 2025, doi: 10.1007/s44196-025-00954-w.

[23] Z. Lou, S. Wang, X. Yu, and W. Song, "Predicting Customer Buying Habits Using Convolutional Neural Network," *Journal of Big Data*, vol. 13, p. 122, 2026, doi: 10.1186/s40537-026-01464-y.

[24] L. Simon, "A Generalised Comparison of Pareto/NBD Based Forecasts Using MCMC, Maximum Likelihood, and Heuristics," *Journal of Business Economics*, vol. 95, pp. 1079–1105, 2025, doi: 10.1007/s11573-025-01237-8.

[25] L. González-Flores, J. Rubiano-Moreno, and G. Sosa-Gómez, "The Relevance of Lead Prioritization: A B2B Lead Scoring Model Based on Machine Learning," *Frontiers in Artificial Intelligence*, vol. 8, p. 1554325, 2025, doi: 10.3389/frai.2025.1554325.

[26] D. O'Brien and R. E. Kunz, "Will They Stay or Will They Go? Application of Machine Learning Classifiers of Purchase and Churn Prediction to a Digital Journalistic Platform," *Journal of Media Business Studies*, vol. 22, no. 3, pp. 237–265, 2025, doi: 10.1080/16522354.2024.2444075.

[27] Ministry of Commerce and Industry, "GeM completes a decade of transforming public procurement with cumulative GMV exceeding Rs. 20 lakh crore," Press Information Bureau (PIB), Delhi, Aug. 7, 2026. [Online]. Available: https://www.pib.gov.in/PressReleasePage.aspx?PRID=2296138 (Accessed on: Oct. 25, 2026).

[28] V. S. Narwane and P. Priyadarshinee, "Examining the effect of AI-BDA on manufacturing firm performance: An Indian approach," *International Journal of Information Management Data Insights*, vol. 5, art. no. 100306, 2025, doi: 10.1016/j.jjimei.2024.100306.

[29] F. Stefano, F. Zanichelli, and D. Jannach, "B2B-Parts-Rec: A Novel Dataset to Study Business-to-Business Recommendation Problems," Zenodo, Apr. 2026, doi: 10.5281/zenodo.19492687. [Online]. Available: https://zenodo.org/records/19492687